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**University of Information Technology & Sciences**

## Project Report

### **E-Commerce Product Search Engine Using Inverted Index and TF-IDF Ranking**

**Course Code:** CSE426

**Course Title:** Data Mining and Warehouse Lab

#### Submitted To

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# **E-Commerce Product Search Engine Using Inverted Index and TF-IDF Ranking**

## **1. Introduction**

### **1.1 Background**

E-commerce websites contain thousands of products distributed across multiple categories. Users rely on search engines to locate desired products efficiently. A search engine must not only retrieve relevant products but also rank them according to their relevance to the user's query.

Modern search systems use indexing techniques and ranking algorithms to improve search speed and accuracy. One of the most widely used indexing methods is the Inverted Index, while TF-IDF remains a popular ranking technique in information retrieval systems

### **1.2 Problem Statement**

Searching through a large product collection using sequential scanning is inefficient and time-consuming. The challenge is to design a search engine that can:

- Retrieve products quickly.
- Handle large datasets efficiently.
- Support Boolean query processing.
- Rank search results based on relevance.
- Provide a user-friendly interface.

### **1.3 Objectives**

The objectives of this project are:

- To build a custom inverted index from product data.
- To implement text preprocessing techniques.
- To support keyword and Boolean searching.
- To rank search results using TF-IDF.
- To create a web-based search interface.
- To deploy the application online.

### **1.4 Scope**

The system focuses on e-commerce product retrieval and ranking. It supports keyword search, Boolean search, ranking, product display, and online access.

## 2. Dataset Description

### 2.1 Dataset Overview

The project uses the Flipkart E-Commerce Product Dataset.

The dataset contains product-related information such as:

- Product Name
- Product Category
- Product Description
- Brand Name
- Retail Price
- Discounted Price
- Product URL
- Product Image URL

The dataset includes thousands of products from different categories including clothing, electronics, footwear, furniture, sports items, and household products.

### 2.2 Dataset Features

The following fields were utilized:

| Attribute        | Description         |
|------------------|---------------------|
| product_name     | Product title       |
| description      | Product description |
| brand            | Brand name          |
| retail_price     | Original price      |
| discounted_price | Discounted price    |
| product_url      | Product page link   |
| image            | Product image URL   |

## 3. Data Preprocessing

### 3.1 Text Cleaning

The raw product data contains various inconsistencies including punctuation, capitalization differences, and irrelevant words. Therefore, text preprocessing was performed before indexing.

### **Lowercase Conversion**

All text was converted into lowercase to ensure case-insensitive matching.

Example: Samsung → samsung

### **Punctuation Removal**

Special symbols and punctuation marks were removed.

### **Tokenization**

Text was split into individual tokens.

Example: "Samsung Galaxy Phone"

Becomes: ["samsung", "galaxy", "phone"]

### **Stopword Removal**

Common words with little semantic meaning were removed.

Examples:

- the
- is
- a
- an
- and
- of

### **Stemming**

Porter Stemmer was used to reduce words to their root forms.

Examples:

- phones → phone
- shoes → shoe
- products → product



Figure: Data Preprocessing Workflow

Dataset → Cleaning → Tokenization → Stopword Removal → Stemming → Processed Dataset

## Chapter 4: Inverted Index Construction

### 4.1 Inverted Index Concept

An inverted index maps terms to the documents in which they appear.

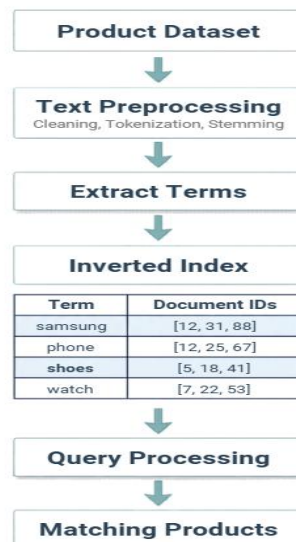


Figure: Structure of Inverted Index

## 4.2 Index Design

Each indexed term stores:

- Document ID
- Term Frequency
- Position Information

## 4.3 Advantages of Inverted Index

- Fast Retrieval
- Reduced Search Complexity
- Scalability
- Efficient Searching

# Chapter 5: Query Processing and Ranking

## 5.1 Query Preprocessing

User queries undergo:

- Lowercasing
- Tokenization
- Stopword Removal
- Stemming

## 5.2 Boolean Query Processing

### AND Query

Example: samsung AND phone

### OR Query

Example: watch OR shoes

## 5.3 TF-IDF Ranking

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

The ranking mechanism prioritizes more relevant products.

## 5.4 Benefits of TF-IDF

- Improves ranking quality
- Reduces effect of common words
- Returns more relevant results

# Chapter 6: System Design and Implementation

## 6.1 System Architecture



Figure: Overall System Architecture

## 6.2 Development Tools

- Python
- Pandas
- NLTK
- JSON
- Gradio
- Hugging Face Spaces

## 6.3 User Interface Design

The image shows the user interface of the ShopSearch application. At the top, there is a header with a shopping cart icon and the text "ShopSearch". Below the header, a subtitle reads "E-Commerce Product Search Engine using Custom Inverted Index and TF-IDF Ranking". The main content area contains a search form with a "Search Product" label, a text input field with placeholder text "Search for samsung phone, shoes or watch...", a "Number of Results" label, and a dropdown menu currently set to "10". A "Search" button is located at the bottom of the form. At the very bottom of the page, there is a footer with the text "Use via API" followed by a small icon, "Built with Gradio" followed by a small icon, and "Settings" followed by a gear icon.

Figure: Search Engine Homepage

## 6.4 Product Result Card

Each card displays:

- Product Name
- Brand
- Retail Price
- Discounted Price
- TF-IDF Score
- Product Image
- Product URL

 **ShopSearch**

E-Commerce Product Search Engine using Custom Inverted Index and TF-IDF Ranking

Search Product

watch

Number of Results

10

Search

Found 579 results in 0.0167 seconds



**Fastrack 9912PP15 Tees Analog Watch - For Men, Women**  
**Brand:** nan  
**Retail Price:** ₹750.0 | **Discounted Price:** ₹750.0  
**TF-IDF Score:** 131.6748  
Fastrack 9912PP15 Tees Analog Watch - For Men, Women Price: Rs. 750 The Fastrack Tees series of watches is the perfect accessory to team up with a pair of blue cotton shorts, a T-shirt and comfortable loafers on your feet when you are on a roa...  
[View Product](#)

Figure: Product Result Card with Image and URL



# Chapter 7: Experimental Results

## 7.1 Query Test 1

Query: samsung phone



**AdroitZ Premium Phone Socket Holder For Samsung Galaxy S5**  
Brand: AdroitZ  
Retail Price: ₹599.0 | Discounted Price: ₹164.0  
TF-IDF Score: 75.0975  
Key Features of AdroitZ Premium Phone Socket Holder For Samsung Galaxy S5 Type: Anti-slip Orientation: Portrait Mount Type Dashboard: Yes,AdroitZ Premium Phone Socket Holder For Samsung Galaxy S5 (Red, Black) Price: Rs. 206 Description:- 1. The clips is adjustable: The max openin...  
[View Product](#)



**AdroitZ Premium Phone Socket Holder For Samsung Infuse 4G**  
Brand: AdroitZ  
Retail Price: ₹599.0 | Discounted Price: ₹164.0  
TF-IDF Score: 75.0975  
Key Features of AdroitZ Premium Phone Socket Holder For Samsung Infuse 4G Type: Anti-slip Orientation: Portrait Mount Type Dashboard: Yes,AdroitZ Premium Phone Socket Holder For Samsung Infuse 4G (Red, Black) Price: Rs. 206 Description:- 1. The clips is adjustable: The max openin...  
[View Product](#)



**Karp Bracelet Necklace Pearl wrist line data portable beaded fashion bracelet USB charging cable for Samsung HTC Phone-Pink USB-P-2811 USB USB Charger**  
Brand: Karp  
Retail Price: ₹1099.0 | Discounted Price: ₹599.0

 **ShopSearch**  
E-Commerce Product Search Engine using Custom Inverted Index and TF-IDF Ranking

Search Product

samsung phone

Number of Results

10

Search

Found 41 results in 0.0229 seconds



**AdroitZ Premium Phone Socket Holder For Samsung ATIV S**  
Brand: AdroitZ  
Retail Price: ₹599.0 | Discounted Price: ₹164.0  
TF-IDF Score: 75.0975  
Key Features of AdroitZ Premium Phone Socket Holder For Samsung ATIV S Type: Anti-slip Orientation: Portrait Mount Type Dashboard: Yes,AdroitZ Premium Phone Socket Holder For Samsung ATIV S (Red, Black) Price: Rs. 206 Description:- 1. The clips is adjustable: The max opening is 8...  
[View Product](#)

Figure: Search Results for Samsung Phone

**Observation:** Relevant Samsung products were retrieved successfully.

## 7.2 Query Test 2

Query: shoes

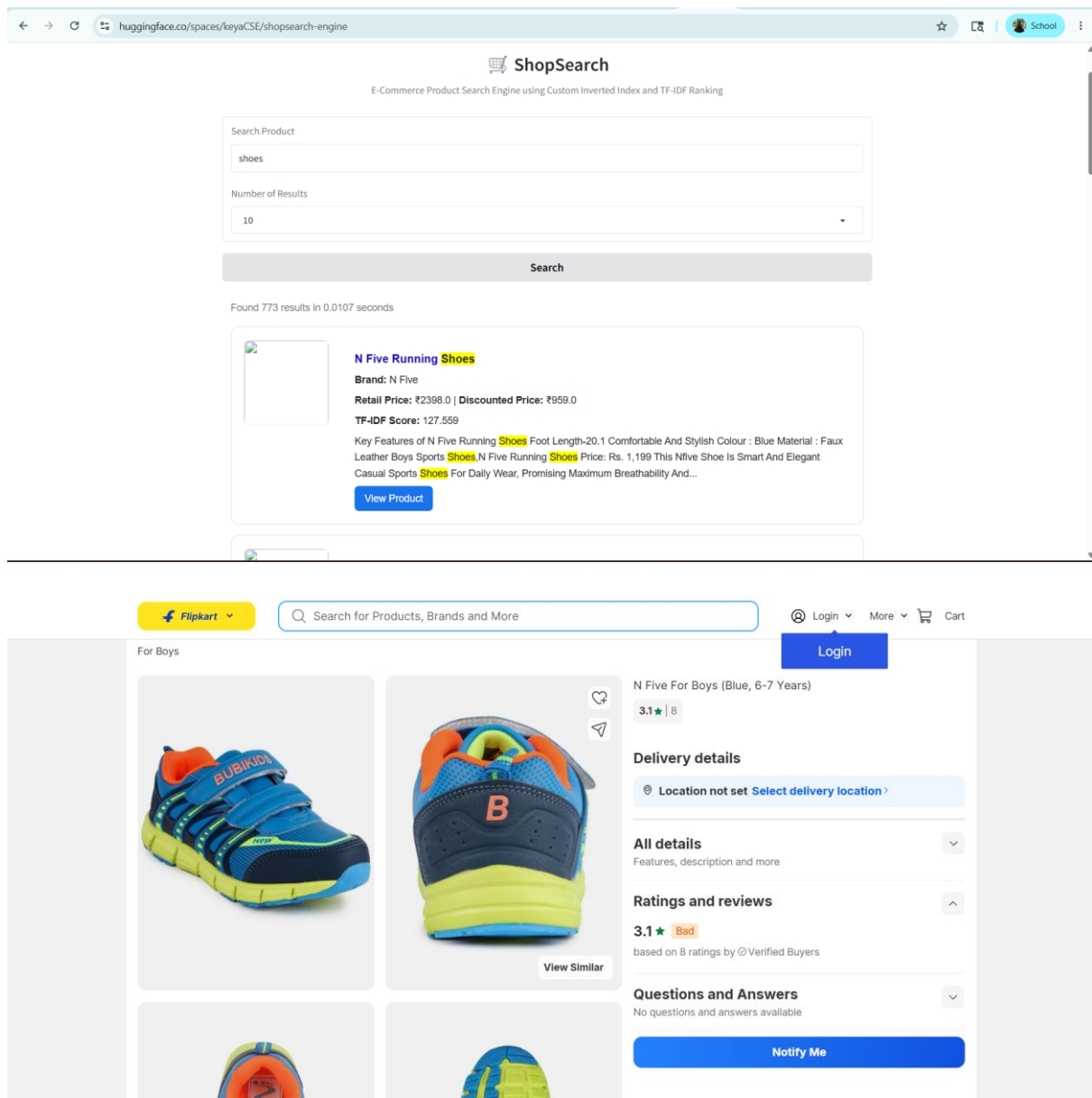


Figure: Search Results for Shoes

**Observation:** Footwear-related products were ranked correctly.

## 7.3 Query Test 3

Query: watch OR shoes

The screenshot displays the ShopSearch interface. At the top, the search bar contains the query "watch OR shoes". Below the search bar, the number of results is set to 10. A "Search" button is visible. The results section shows "Found 1351 results in 0.1333 seconds". The first result is "Fastrack 9912PP15 Tees Analog Watch - For Men, Women" with a brand of "nan", a retail price of ₹750.0, and a discounted price of ₹750.0. The TF-IDF score is 131.6748. The description mentions "The Fastrack Tees series of watches is the perfect accessory to team up with a pair of blue cotton shorts, a T-shirt and comfortable loafers on your feet when you are on a roa...". A "View Product" button is present. The second result is "N Five Running Shoes" with a brand of "N Five", a retail price of ₹2398.0, and a discounted price of ₹959.0. The TF-IDF score is 127.559. The description mentions "Foot Length-20.1 Comfortable And Stylish Colour : Blue Material : Faux Leather Boys Sports Shoes, N Five Running Shoes Price: Rs. 1,199 This Nfive Shoe Is Smart And Elegant Casual Sports Shoes For Daily Wear, Promising Maximum Breathability And...". A "View Product" button is present. The third result is "Anno Dominii ADWB0000139 Watch Box" with a brand of "Anno Dominii", a retail price of ₹1499.0, and a discounted price of ₹1499.0. The TF-IDF score is 118.0532. The description mentions "Box 5 Separate cushioned compartments for watches High Quality Leatherette 5 removable Cushions AD Watch Case, Anno Dominii ADWB0000139 Watch Box (Black, Holds 5 Watches) Price: Rs. 1,499 Anno Dominii Leatherette Designer watch box for...". A "View Product" button is present. The fourth result is "V-Luma W600 Fashion Analog Watch - For Men" with a brand of "V-Luma", a retail price of ₹1599.0, and a discounted price of ₹699.0. The TF-IDF score is 118.0532. The description mentions "Box 5 Separate cushioned compartments for watches High Quality Leatherette 5 removable Cushions AD Watch Case, Anno Dominii ADWB0000139 Watch Box (Black, Holds 5 Watches) Price: Rs. 1,499 Anno Dominii Leatherette Designer watch box for...". A "View Product" button is present.

ShopSearch  
E-Commerce Product Search Engine using Custom Inverted Index and TF-IDF Ranking

Search Product  
watch OR shoes

Number of Results  
10

Search

Found 1351 results in 0.1333 seconds

**Fastrack 9912PP15 Tees Analog Watch - For Men, Women**  
Brand: nan  
Retail Price: ₹750.0 | Discounted Price: ₹750.0  
TF-IDF Score: 131.6748  
Fastrack 9912PP15 Tees Analog Watch - For Men, Women Price: Rs. 750 The Fastrack Tees series of watches is the perfect accessory to team up with a pair of blue cotton shorts, a T-shirt and comfortable loafers on your feet when you are on a roa...  
View Product

**N Five Running Shoes**  
Brand: N Five  
Retail Price: ₹2398.0 | Discounted Price: ₹959.0  
TF-IDF Score: 127.559  
Key Features of N Five Running Shoes Foot Length-20.1 Comfortable And Stylish Colour : Blue Material : Faux Leather Boys Sports Shoes, N Five Running Shoes Price: Rs. 1,199 This Nfive Shoe Is Smart And Elegant Casual Sports Shoes For Daily Wear, Promising Maximum Breathability And...  
View Product

**Anno Dominii ADWB0000139 Watch Box**  
Brand: Anno Dominii  
Retail Price: ₹1499.0 | Discounted Price: ₹1499.0  
TF-IDF Score: 118.0532  
Key Features of Anno Dominii ADWB0000139 Watch Box 5 Separate cushioned compartments for watches High Quality Leatherette 5 removable Cushions AD Watch Case, Anno Dominii ADWB0000139 Watch Box (Black, Holds 5 Watches) Price: Rs. 1,499 Anno Dominii Leatherette Designer watch box for...  
View Product

**V-Luma W600 Fashion Analog Watch - For Men**  
Brand: V-Luma  
Retail Price: ₹1599.0 | Discounted Price: ₹699.0  
TF-IDF Score: 118.0532  
Key Features of V-Luma W600 Fashion Analog Watch 5 Separate cushioned compartments for watches High Quality Leatherette 5 removable Cushions AD Watch Case, Anno Dominii ADWB0000139 Watch Box (Black, Holds 5 Watches) Price: Rs. 1,499 Anno Dominii Leatherette Designer watch box for...  
View Product

Figure: OR Query Result

**Observation:** Products containing either keyword were returned.

## 7.4 Query Test 4

**Query:** samsung AND phone

ShopSearch

E-Commerce Product Search Engine using Custom Inverted Index and TF-IDF Ranking

Search Product

samsung AND phone

Number of Results

20

Search

Found 41 results in 0.0017 seconds

AdroiZ Premium Phone Socket Holder For Samsung ATIV S

Brand: AdroiZ

Retail Price: ₹599.0 | Discounted Price: ₹164.0

TF-IDF Score: 75.0975

Key Features of AdroiZ Premium Phone Socket Holder For Samsung ATIV S Type: Anti-slip Orientation: Portrait Mount Type Dashboard: Yes, AdroiZ Premium Phone Socket Holder For Samsung ATIV S (Red, Black) Price: Rs. 206 Description:- 1. The clips is adjustable. The max opening is 8...

99Gems LR GOLD 5 in 1 MOBILE / MP3 USB Cable

Brand: 99Gems

Retail Price: ₹199.0 | Discounted Price: ₹149.0

TF-IDF Score: 50.3929

99Gems LR GOLD 5 in 1 MOBILE / MP3 USB Cable (black) Price: Rs. 149 Sure To Match Your Fashion Electronics, This Black In Color 5 in 1 Data Cable From The Latest Collection , A Great Pick Against Your Valuable Currency. Perfectly Compatible Wi...

View Product

Shop Swipe Anti-Radiation Retro Style Wired Headphones

Brand: Shop Swipe

Retail Price: ₹1200.0 | Discounted Price: ₹559.0

TF-IDF Score: 50.3929

Key Features of Shop Swipe Anti-Radiation Retro Style Wired Headphones Headset Design: Canalphone Type: Over the Ear Compatible Devices: Mobile, Tablet, Computer, Shop Swipe Anti-Radiation Retro Style Wired Headphones (Black, Over the Ear) Price: Rs. 559 Shop Swipe Wired Handset S...

View Product

EDGE PLUS BODY PANEL FOR SAMSUNG S3500 Front & Back Panel

Brand: EDGE PLUS

Retail Price: ₹1199.0 | Discounted Price: ₹599.0

Figure: AND Query Result

**Observation:** Only products containing both keywords were returned.

## 7.5 Performance Analysis

The system demonstrated:

- Fast Retrieval Time
- Efficient Query Processing
- Accurate Ranking
- Responsive Interface

## Chapter 8: Online Deployment

### 8.1 Deployment Platform

The application was deployed using Hugging Face Spaces.

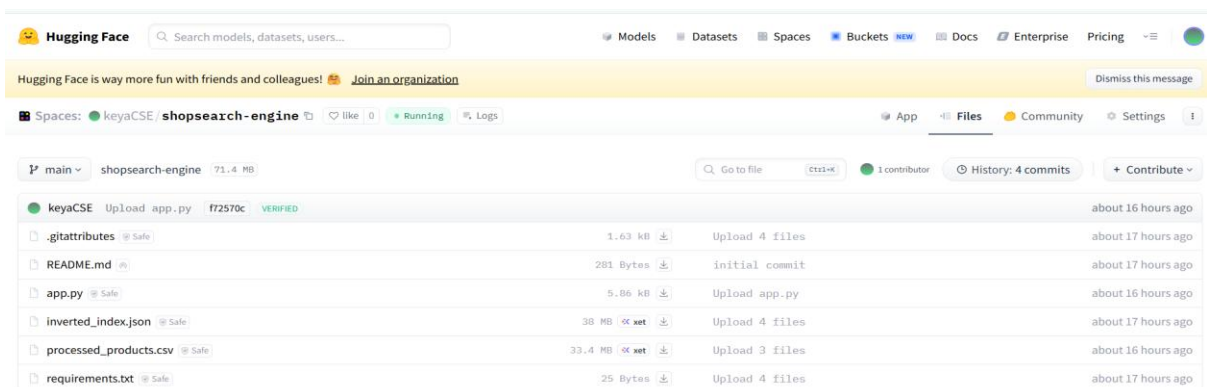


Figure 8.1: Deployed Search Engine on Hugging Face

### 8.2 Benefits of Deployment

- Accessible Anywhere
- No Installation Required
- Easy Demonstration
- Real-Time Searching

## Chapter 9: Advantages

- Custom Inverted Index
- TF-IDF Ranking
- Boolean Query Support
- Product Image Display
- Product URL Support
- Fast Retrieval
- Online Deployment

## **Chapter 10: Limitations**

- Some image links may be unavailable.
- No semantic search support.
- No spelling correction.
- No recommendation system.

## **Chapter 11: Future Work**

- Semantic Search
- Spell Correction
- Auto Completion
- Personalized Recommendation
- Deep Learning Ranking
- Voice Search

## **Chapter 12: Conclusion**

The project successfully developed a specialized E-Commerce Product Search Engine using Inverted Index and TF-IDF ranking techniques. The system efficiently retrieves relevant products, supports Boolean queries, displays product information with images and URLs, and is deployed online using Hugging Face Spaces. Experimental results demonstrate the effectiveness of the proposed search engine in handling large-scale e-commerce product collections.

[Link: https://huggingface.co/spaces/keyaCSE/shopsearch-engine](https://huggingface.co/spaces/keyaCSE/shopsearch-engine)

[GitHub: https://github.com/keyaCSE/CSE426](https://github.com/keyaCSE/CSE426)